USER MANUAL AND ACRONYMS

User Manual

This system enables users to detect Autism Spectrum Disorder (ASD) from preprocessed resting-state fMRI data. It classifies patients and visualizes important brain regions in both static and dynamic 3D formats.

Fig 1: Landing Page of ASD Detection software

Steps to Use the System:

1. **Enter File ID:**
   a. The user begins by entering a valid File_ID corresponding to a patient's preprocessed fMRI dataset.
   b. The system verifies and loads the associated data.
2. **Upload and Model Processing:**
   a. Upon entering the File ID, the system automatically uploads the linked data and processes it through a trained classification model.
   b. The model outputs a prediction: either **ASD (Autism Spectrum Disorder)** or **TC (Typically Developed Control)**.

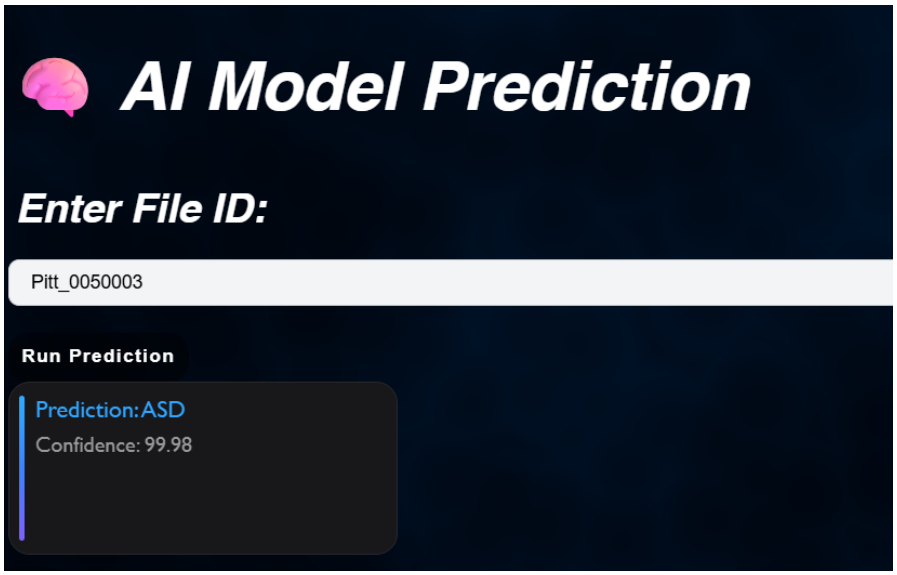

Fig 2: AI Model Prediction

3. **3D Brain Visualization:**

   a. After classification, the system provides two types of visualizations:

      i. **Static Visualization:**

         1. Displays important **Regions of Interest (ROIs)** identified as contributing to the diagnosis.
         2. The user can interact with the brain model by clicking or hovering over regions to see names or importance scores.

      ii. **Dynamic Visualization:**

         1. Animates how activity within the selected important ROIs changes over the time series.
         2. Offers insights into temporal brain dynamics associated with ASD or typical patterns.

4. **Interpreting the Results:**

   a. The system visually emphasizes brain areas relevant to the decision.

b. Users can explore how specific ROIs behave differently in ASD compared to TC cases.

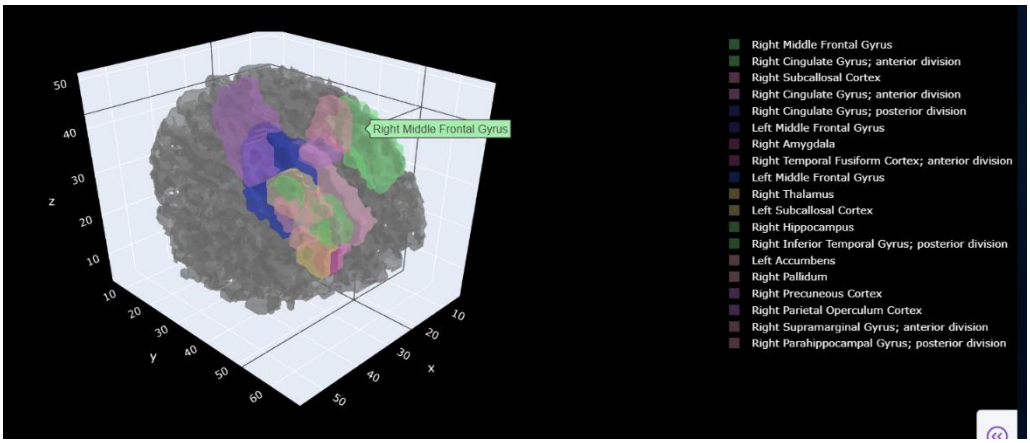

Fig 3: Visualization of Important ROI's for ASD Detection

5. **System Requirements:**

a. Ensure the input file corresponds to valid preprocessed fMRI data.
b. A stable internet connection and modern browser are recommended for 3D interactions.

Acronyms

Acronym	Full Form
ASD	Autism Spectrum Disorder
TC	Typical Controls
fMRI	Functional Magnetic Resonance Imaging
Rs-fMRI	Resting State - fMRI
ROI	Region of Interest
GCN	Graph Convolutional Network
MLP	Multi-Layer Perceptron
GUI	Graphical User Interface
AUROC	Area Under the Receiver Operating Characteristic Curve
BOLD	Blood Oxygen Level Dependent (signal)
MRI	Magnetic Resonance Imaging
ReLU	Rectified Linear Unit (activation function)
DTI	Diffusion Tensor Imaging
HO	Harvard-Oxford Atlas
ABIDE	Autism Brain Imaging Data Exchange
PC	Pearson Correlation